



**CLEAN DEVELOPMENT MECHANISM
PROJECT DESIGN DOCUMENT FORM (CDM-PDD)
VERSION 02 - IN EFFECT AS OF: 1 JULY 2004)**

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**SECTION A. General description of project activity****A.1 Title of the project activity::**

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La Higuera Hydroelectric Project, Chile

Version number: Version 8

Date: 6 January 2012

A.2. Description of the project activity:

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Purpose of project activity

The La Higuera Project (hereafter, the Project) is being developed by Hidroeléctrica La Higuera S.A. (HLH), a joint venture between Pacific Hydro Limited (PHL), through its subsidiary Pacific Hydro Chile S.A (PHC), and Statkraft Norfund Power Invest (SNPI). The purpose of the project is to utilise the hydrological resources of the Tinguiririca and Azufre Rivers in a run of river scheme to generate zero emission energy to the Chilean Central electricity grid (SIC). The Project will generate certified emission reductions (CERs) by displacing electricity generation from grid connected fossil fuel-fired power plants that would otherwise be generating electricity.

In May 2001 PHC secured an option to purchase the water rights. It has taken 2 years of intense negotiation to finalise a Power Purchase Agreement, which was only executed in June 2005. In May 2004 a power purchase agreement (PPA) was signed in principle with SAESA, the forth-largest distributor in Chile with concessions in the southern portions of SIC. However, negotiations on this contract could not be finalised, mainly due to the financing requirements of the Projects' potential Lenders not being acceptable to SAESA. In May 2004 PHL sold down part of its ownership of the project by 50% to SNPI. In August 2004 the Project was granted Environmental Approval (EIA) by the Chilean Government. In February 2005 Host Country Approval was granted by the Chilean DNA. Based on annual estimated generation of 811 GWh the project will reduce CO₂ emissions by an estimated 405,446 tonnes per annum.

The Project will contribute to the social welfare of the 6th region, Chile, where local employment opportunities and infrastructure are poor. Construction of the Project will last 30 months, with the Contractor required to source at least 30% of its workforce locally. Road access to the valley and its upper reaches is limited to a few months in summer due to the treacherous nature of the road. Road access will be improved to year-round access, opening up the area to income generating activities including tourism, to fully utilise the natural resources such as thermal springs, climbing and other recreational activities.

Contribution to sustainable development

As mentioned above, host country approval for the project issued by the Chilean DNA has already been received. The Designated National Authority checked the project against the sustainable development criteria of the country, finding that La Higuera project contributes towards meeting those.

The Project is also particularly relevant to the sustainable development of the Chilean electricity sector, which is currently undergoing a fuel supply crisis. Following the introduction of imported Argentinean natural gas in 1996, most additional capacity needs have been met by combined cycle natural gas plants.

However, in 2004 Argentina restricted the exports of natural gas into Chile and subsequent gas restrictions have forced many of these plants to reduce generation and use diesel where environmental permissions exist. As a result of this the Chilean Government is actively seeking alternative forms of energy to supply a system with an internal growth demand of 350-400 MW per annum. Unfortunately, renewable energies are not a priority, favouring again more thermal energy like coal fired plants. The role of the project in contributing to the sustainable development of Chile and its additionality in such a scenario is a clear demonstration of the value of the Project.

**A.3. Project participants:**

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Name of Party involved ((host) indicates a host Party)	Private and/or public entity(ies) project participants (as applicable)	Kindly indicate if the Party involved wishes to be considered as project participant (Yes/No)
Chile (host)	Hidroeléctrica La Higuera S.A	NO

Table 1 - Project participants

Contact details information on project participants are provided in Annex 1.

A.4. Technical description of the project activity:**A.4.1. Location of the project activity:**

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The Project will be located in the valley of the Tinguiririca River. This river rises in the Andes Mountains about 120km south of Santiago and flows westward to the Pacific Ocean. The nearest major city to the project area is San Fernando. The area of the Project activity is shown in Figure 1 below.

**Figure 1** - Location of Project Activity**A.4.1.1. Host Party(ies):**

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Chile

A.4.1.2. Region/State/Province etc:

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6th Administrative Region, Chile

A.4.1.3. City/Town/Community etc:

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The nearest community to the Project is Puente Negro, approximately 15kms downstream from the powerhouse. The nearest city is San Fernando, which is on the PanAmerican Highway, 120 km south of the capital city, Santiago.

A.4.1.4. Detail of physical location, including information allowing the unique identification of this project activity (maximum one page):

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The Project is located in the Tinguiririca Valley, specifically between the elevations of 1100 m.a.s.l. to 728 m.a.s.l as shown in Figure 2.

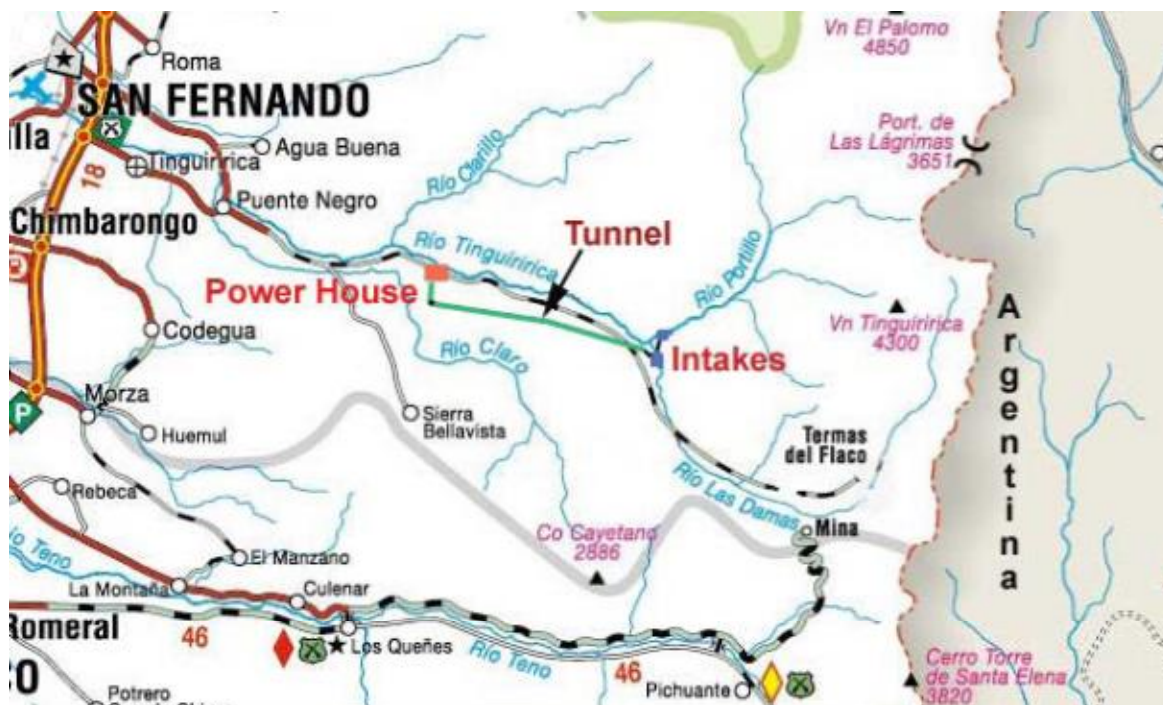


Figure 2 - Location of Project Activity

A.4.2. Category(ies) of project activity:

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Sectoral Category 1, Energy Industries (renewable/non renewable)

A.4.3. Technology to be employed by the project activity:

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The project considers an intake on Los Helados River that discharge its water into the Azufre River, immediately upstream of the Azufre intake, at 1104 m.a.s.l, diverting partial flows via a canal and tunnel to the Tinguiririca intake. Water from the Tinguiririca River, including the water from the Azufre intake and Los Helados intake, is diverted via a weir at 1100 m.a.s.l. into desanding bays on the north bank of the Tinguiririca. This water is then conveyed some 18km in a 5.8m diameter tunnel with a design flow of 50 m³/s into an off channel reservoir of 150,000m³ capacity. Water is conveyed through a tunnel and



steel pipe from this off channel head pond to a surge shaft and valve chamber, where it enters a surface steel penstock of 600m length. The La Higuera powerhouse is located on the south bank of the Tinguiririca at around 715 m.a.s.l.

The tailrace returns the diverted flows to the river approximately 100m downstream of the powerhouse. Electricity is transmitted approximately 38km along a 154kv high tension transmission line to a substation near the town of San Fernando, where it joins the SIC grid.

The Project will use state of the art but known technology in electricity generation and transmission. In particular the vertical axis Francis turbines will be provided by Power Machines (Russian company), generators by Electrosila and transformers to be appointed. Both companies have been utilising these forms of technology for a number of decades, while control and safety systems are state of the art. Specifically this is the first application of Francis turbines in a high head project in Chile.

La Higuera Project		
Gross head	385m	
Machine type	2 x 77.5MW Francis	
Design Flow	50m ³ /s	
Installed Capacity	155 MW	
Annual average Generation	811 GWh	
Storage facilities	Surface area	Capacity
Off channel	22,500 m ²	150,000 m ³

A.4.4 Brief explanation of how the anthropogenic emissions of anthropogenic greenhouse gas (GHGs) by sources are to be reduced by the proposed CDM project activity, including why the emission reductions would not occur in the absence of the proposed project activity, taking into account national and/or sectoral policies and circumstances

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How the emissions are reduced

La Higuera hydro plant will reduce emissions by displacing electricity generated from fuel-based power plants. La Higuera hydro plant is a run-of-river with no project emissions associated with its operations. The current structure of the SIC is such that the generation mix capacity comprises of 60% hydroelectric generation, 30% natural gas, and the remainder from coal, diesel, petcoke and cogeneration. The coal, petcoke, petroleum and diesel plants have the highest marginal costs, and as the SIC uses least cost order of merit dispatch these plants are typically the marginal operating plants in the system. Any run of river plant, with zero marginal cost, as such displaces these plants to the extent of its generation when these plants would have otherwise been dispatched. As such there is a clearly demonstrable reduction in GHG emissions from the introduction of La Higuera into the SIC system.

Why the emission reductions would not occur otherwise

La Higuera will be the second biggest run of river project ever built by installed capacity in Chile¹. In 1991 the 160 MW Alfalfal plant was built on the Colorado River, some 80km north of the La Higuera project. This project is similar in many ways to La Higuera project, however, the economics of the Alfalfal project were significantly better for two main reasons. Firstly, in 1991 there was no gas available, and as such node prices for energy were significantly higher than current prices. The second factor is that the owners, GENER, also have coal fired thermal plants, allowing them to mitigate the high seasonal generation of low winter generation and high summer snow melt generation.

¹ This does not include run of river plants associated with large reservoirs, which provide annual regulation, and as such should not be considered as run of river projects.



Following the introduction of imported natural gas from Argentina in 1996 there was significant change in the electricity sector's structure, with high growth in the lower capital cost and shorter payback period combined cycle natural gas plants as shown in Figure 3. This resulted in a significant reduction in node prices. This has increased the barriers to hydroelectricity and other renewable energy technologies, which are typically much higher capital cost investments.

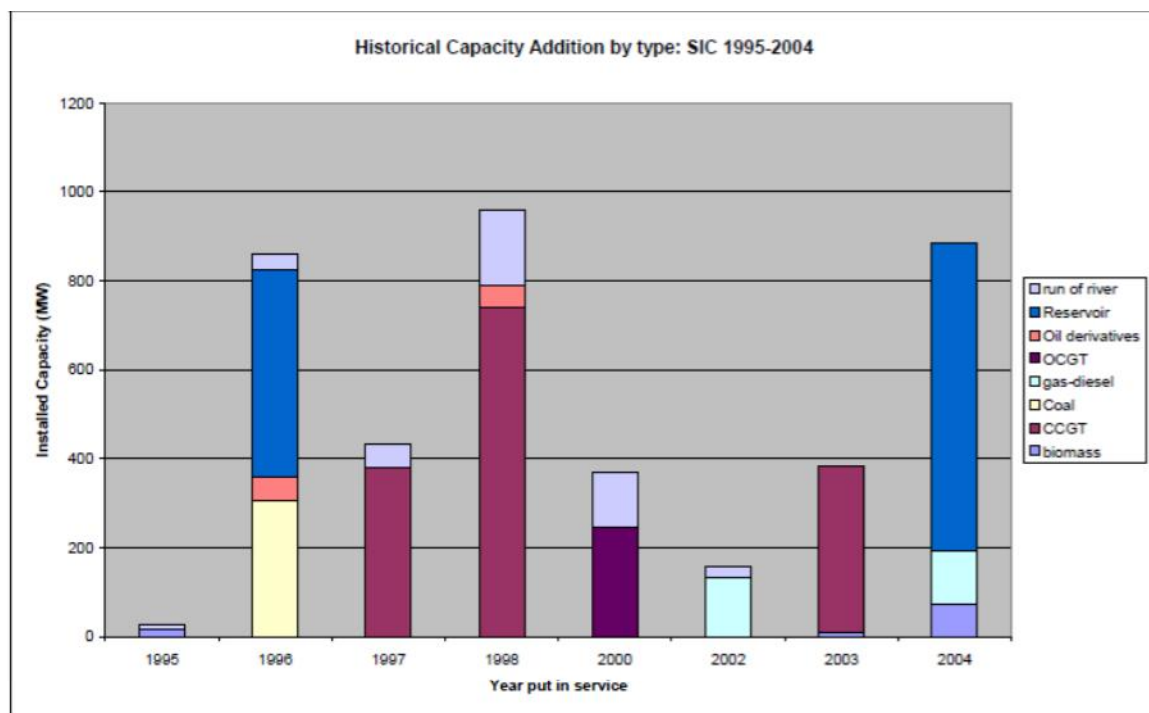


Figure 3 - Historical Capacity Addition By type SIC 1995 – 2004

Total emissions reductions:

In total, the project is estimated to reduce emissions of greenhouse gases by 2,838,122 tCO₂e over a period of 7 years.

A.4.4.1 Estimated amount of emission reductions over the chosen crediting period:

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The project will displace electricity from a more carbon-intensive grid with an estimated combined margin emission factor of 0.501 tCO₂/MWh. The project is expected to displace 811 GWh of electricity per year, thus reducing GHG emissions by 405,446 tCO₂e per year in the baseline scenario.

Years	Annual estimation of emission reductions in tonnes CO ₂ e
1	405,446
2	405,446
3	405,446
4	405,446
5	405,446
6	405,446
7	405,446
Total estimated reductions (tonnes of CO ₂ e)	2,838,122



Total number of crediting years	7
Annual average over the crediting period of estimated reductions (tonnes of CO ₂ e)	405,446

Table 2 - Estimation of emission reductions in tonnes of CO₂e for the crediting period

Refer to section E for further details on the quantification of GHG emission reductions associated with this project.

A.4.5. Public funding of the project activity:

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There is no public funding being provided to the project.

**SECTION B. Application of a baseline methodology****B.1. Title and reference of the approved baseline methodology applied to the project activity:**

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ACM0002: “Consolidated baseline methodology for grid-connected electricity generation from renewable sources”.

B.1.1. Justification of the choice of the methodology and why it is applicable to the project activity:

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The project activity is a grid connected run-of-river hydropower project, where the grid’s geography and system boundaries are explicit and characteristics are readily available through CNE and CDEC-SIC. As a renewable energy project it is appropriate to follow Paragraph 48 of Marrakech Accords and use existing actual or historical emissions, since the project activity will serve to reduce actual emissions. On this basis the conditions for applying ACM0002 are met.

B.2 Description of how the methodology is applied in the context of the project activity:

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Applying the methodology ACM0002 “Consolidated baseline methodology for grid-connected electricity generation from renewable sources”, the implementation of the La Higuera project will reduce on average 405,446 tCO₂ per annum.

As per methodology, the baseline scenario is the “Electricity delivered to the grid by the project would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described below.

Emission factors for each grid-connected power plant are calculated by using the corresponding electricity generation on an hourly basis data provided by the CDEC-SIC and the fuel consumption per year provided on a yearly publication by the same entity (CDEC-SIC) . This information allows the determination of an ex post baseline Emission Factor for the project activity.

The baseline emission factor (EF_y) is calculated as a combined margin (CM), consisting of the combination of operating margin (OM) and build margin (BM) factors according to the following three steps. Calculations for this combined margin are based on data from an official source and IPCC.

The calculation of the Operating Margin emission factor(s) ($EF_{OM,y}$) has to be based on one of the four following methods:

- (a) Simple OM, or
- (b) Simple adjusted OM, or
- (c) Dispatch Data Analysis OM, or
- (d) Average OM.

Dispatch Data Analysis should be the first methodological choice. Since the fuel consumption - Parameter 5 $F_{i,y}$ as per methodology ACM0002 version 4 - is not available measured on an hourly basis, but only the yearly specific fuel consumption per plant is reported², the Dispatch Data Analysis is not applicable and

² The information provided by the official source CNE is the specific fuel consumption on an annual basis only; hourly fuel consumption is not reported and monitored.



alternatively Simple Adjusted OM may be used. In absence of the Dispatch Data Analysis option the Simple Adjusted OM is deemed the most accurate method to calculate the OM.

In Chile low-cost resources constitute more than 50% of the total generation of the national grid, which prevents from using option (a) Simple OM. Option (d) Average OM can only be used where low-cost/must run resources constitute more than 50% of total grid generation and data to apply Option (b) Simple adjusted OM and detailed data to apply Option (c) is not available. As a result, Option (b) Simple adjusted OM is the preferred method. The data used for the calculation of build margin is shown in Annex 3 of this document. The main source of data is CDEC-SIC.

Emission Reductions are calculated with the following equation:

$$ER_y = BE_y - PE_y - L_y$$

Where:

ER: Emission reductions in the year *y* (t CO₂e)

BE: Baseline emissions in the year *y* (t CO₂e)

PE: Project Emissions in the year *y* (t CO₂e)

L: Leakage emissions in the year *y* (t CO₂e)

y: a given year

ACM0002 does not require calculation of project emissions in case of hydro power projects. However, for reasons of conservativeness, project emissions from gen-sets that will be operated in case of grid failures are taken into consideration as follows:

During the construction and the start of the operation of the plant 4 gen-sets have been installed. During operations these gen-sets will be used only to a limited extent as back-up and for emergencies:

- One gen-set in the Power House with 0.599 MW of installed capacity, which will only be used for that specific purpose and will not supply power to the grid.
- One gen-set at the Los Helados intake, with 0.022 MW of installed capacity,
- One gen-set at the Azufre intake with 0.080 MW of installed capacity; and
- One gen-set at the Tinguiririca intake with 0.100 MW of installed capacity, which is used to move the gates, in the event that connection to the grid is lost

It is estimated that emissions from the gen-sets are well below 1% of the average emission reduction of the project activity.

Nevertheless, La Higuera will use hourmeters of the gen-sets in order to be able to monitor the use of the engines in case of a complete grid failure. Related emissions would be considered in the monitoring report in specific verification period.

The emissions of the 4 gen-set correspond to 460 tCO₂. Since the project activity is a renewable energy project consists in hydroelectricity generation, there are no other project emissions.

According to the ACM0002 v.4 methodology project participants do not need to consider leakage for this project activity, then LE_y = 0.

*Baseline Emissions:*

The baseline emission factor calculations will be based on the combined margin using the “Simple Adjusted Operating Margin”, option (b) of the ACM0002, according to the following formulae:

$$BE_y = EG_y * EF_y$$

Where:

BE : Baseline emissions (t CO₂e)

EG_y : Electricity supplied by the project to the grid in the year y (MWh)

EF : baseline emission factor (tCO₂e / MWh)

y : a given year

Where:

$$EF_y = \omega_{OM} * EF_{OM_y} + \omega_{BM} * EF_{BM_y}$$

Where:

EF : baseline emission factor (tCO₂e / MWh)

ω_{OM} : Operation Margin weight, which is 0.5 by default

EF_{OM} : Operational Margin emission factor (tCO₂e / MWh)

ω_{BM} : Build Margin weight, which is 0.5 by default

EF_{BM} : Build Margin emission factor (tCO₂e / MWh)

y : a given year

The Simple Adjusted OM has been chosen and is applied for the project activity ex-post as follows.

(b) *Simple Adjusted OM*. This emission factor ($EF_{OM, simple \ adjusted, y}$) is a variation on the Simple OM method, where the power sources (including imports) are separated in low-cost/must-run power sources (k) and other power sources (j):

$$EF_{OM, simple \ adjusted, y} = (1 - \lambda_y) \cdot \frac{\sum_{i,j} F_{i,j,y} \cdot COEF_{i,j}}{\sum_j GEN_{j,y}} + \lambda_y \cdot \frac{\sum_{i,k} F_{i,k,y} \cdot COEF_{i,k}}{\sum_k GEN_{k,y}}$$

Where $F_{i,j,y}$, $COEF_{i,j}$ and GEN_j are analogous to the variables described for the Simple OM method for plants j :

The CO₂ emission coefficient $COEF_i$ is obtained as:

$$COEF_i = NCV_i \times EF_{CO_2i} \times OXID_i$$

Where

NCV_i is the net calorific value per mass or volume of fuel type i

$OXID_i$ is the oxidation factor of the fuel i ; and

EF_{CO_2} is the CO₂ emission factor per unit of energy of fuel i .

The Build Margin is determined ex-ante as follows:

$$EF_{BM_y} = \frac{\sum_{i,m} F_{i,m,y} * COEF_{i,m}}{\sum_m GEN_{m,y}}$$



Where:

EF_{BM} : Build Margin emission factor (tCO₂e / MWh) and

m : refers to last additions power sources delivering electricity to the grid.

Project Emission:

Emissions of each gen-set is obtained based on metering of the operating hours through the calculation of diesel consumption, the calorific value and the emission factor of the diesel. The default values of NCV and emission factor of the diesel will be derived from the IPCC, higher value at 95% confidence level.

$$PE = \sum FC_i * NCV * EF$$

Where:

FC_i : Diesel consumption of gen-set i

NCV: Net Calorific Value of diesel (default value of 0.0433 TJ/t IPCC, higher value at 95% confidence level)

EF: Diesel emission factor, (default value of 74.8 tCO₂/TJ of IPCC, i.e. higher value at 95% confidence level)

Where the fuel consumption is calculated as

$$FC_i = \text{Hours of operation of gen-set} * \text{specific diesel consumption at full load}$$

Further details are provided in Section D.

The following table provides the key information and data used to determine the baseline scenario:

Variable	Unit	Data Source
Estimated Simple Adjusted Operating Margin Emissions Factor as per 2010 (EF_{OMy} in tCO ₂ /MWh)	0.640	Calculated ex-post using data from CDEC-SIC, CNE and IPCC.
Build Margin Emissions Factor (EF_{BMy} in tCO ₂ /MWh)	0.361	Calculated ex-ante using data from CDEC-SIC, CNE and IPCC.
Baseline Emissions Factor (EF_y in tCO ₂ /MWh)	0.501	Calculated using data from CDEC-SIC, CNE and IPCC.
Electricity generated by Project (EG_y MWh)	811,000	Hidroeléctrica La Higuera S.A
Baseline Emissions (BE tCO ₂)	405,906	Calculated

Table 3 - Data used to determine the baseline scenario

B.3. Description of how the anthropogenic emissions of GHG by sources are reduced below those that would have occurred in the absence of the registered CDM project activity:

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The determination of project scenario additionality is done using the CDM consolidated tool for demonstration of additionality, which follows the following steps

**Step 0. Preliminary screening based on the starting date of the project activity**

The project is only expected to start construction in the first quarter of 2006, which is after November 18 2004 (date of registration of first CDM project), thus crediting period will start at the point of registration of the project. In any case, as it will be demonstrated in the following steps, CDM revenue has been considered from the early stages of development of the project, and it is an integral part of the financial package of the project.

Step 1. Identification of alternatives to the project activity consistent with current laws and Regulations**Sub-step 1a. Alternatives to the project activity**

Alternative options to the project activity include:

- run of river project without CDM assistance;
- natural gas combined; and
- open cycle plants (coal and diesel plants).

Sub-step 1b. Enforcement of applicable laws and regulations

Under the Chilean Electricity Law all the alternatives to the project activity are in compliance with applicable legal and regulatory requirements. No special benefits are established under Chilean laws for renewable energies

Step 3. Barrier Analysis

Following the guidelines for determining additionality there are two analytical options available: investment analysis and barrier analysis. Due to the nature of the project and the important barriers it faces barrier analysis is used. The following barriers have been identified as the crucial barriers affecting the owners in undertaking the project activity.

Sub-step 3a. Identify barriers that would prevent the implementation of type of the proposed project activity:

Hydrological risk:

Hydrology risk is perhaps the greatest risk facing the project as a run of river project with only 5 hours of storage capacity. This capacity allows for storage of 5 hours of design flows discharge during the period of lowest inflows. On this basis during the winter months of April to August the storage would only be sufficient to ensure that the plant could operate for 5 hours each day via discharging the storage and replenishing it for the following day's generation. As a result there is no way the project can mitigate against drought and low inflow periods. If large dams were built on both the Tinguiririca and Azufre Rivers then high summer inflows, where available water in the rivers is generally in excess of capacity of the project, could provide for monthly storage and discharge of the reservoir when the system has less water. Due to almost 100% exposure to meteorological and hydrological risk project returns are highly variable. Figure 4 shows the annual generation the project would have generated based on historical hydrology.

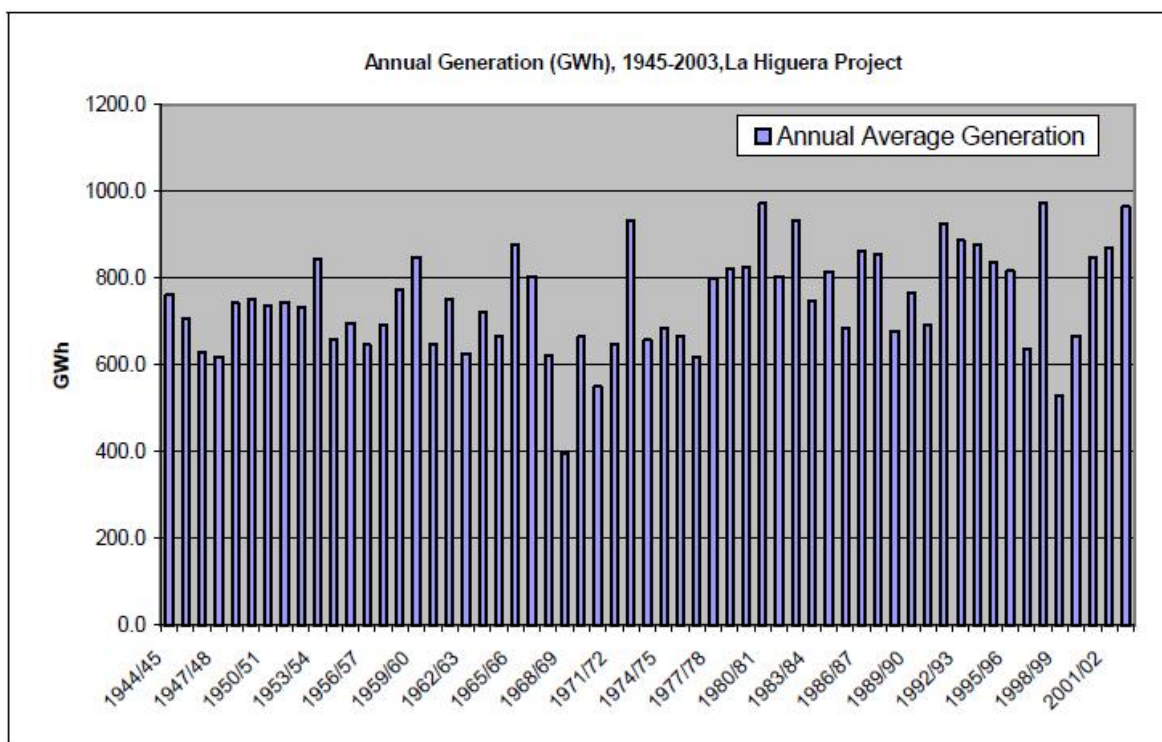


Fig 4 Source: EPS Ingienera

Tunnelling risk:

The project has 17km of tunnels to convey water from the intake to the surge shaft and penstock facility, where it is taken to the powerhouse. Because of the very steep alpine terrain, it is not possible to convey the water through canals, which are much cheaper than tunnels. As a result, the rock conditions for tunnelling are critical to final project cost. The construction company can only rely on surface mapping to base their cost estimates on, and hence there are considerable unknowns. It is not practical to drill sections of the tunnel from the surface, as this information is insufficient for 17km of section. As such risks such as underground water, geotechnical quality and presence of gases have to be taken by the Owner, and not the contractor.

The Owner's Engineer, SwedPower/Golder Associates, have projected an expected worst case overrun of \$3.8m on tunnelling costs. However, the Lender's Independent Engineer, Montgomery Watson Harza, have estimated a more pessimistic scenario of \$6m in possible tunnelling overruns. On this basis the Project has had to increase the contingency for the project. This amount formed part of the initial expected contingency of \$12m. However, in light of this increase and other increases the Project has had to increase the Project Contingency to \$20m to satisfy the Lenders. Delays in tunnelling also expose the Project to completion risk, since the contractor is allowed extensions of time for *force majeure* events, such as encountering underground gases, high water inflows and other natural events. This is industry standard and cannot be avoided. The failure to complete the project on time could make the Project liable under the power supply agreement they have had to sign to obtain financing for the project (see below).



Land Risk:

the project area is sparsely populated during summer time grazing periods, there are 11 landowners in the immediate project area, and another 60 along the route of the transmission line. These landowners have considerable power in extracting payments from the project above market rates to avoid delays. The project developer could try and enforce sale under the Chilean Electricity Law code, but this is a drawn out procedure and does not foster the relationship with stakeholders the company is seeking. In particular the project construction timetable of 30 months assumes that access is to be completed to 2 intermediate adits along the route of the tunnel that would provide additional faces critical for completing the drill and blasting of the tunnels. To settle this issue the Project had to pay over market rates for the land and provide services such as telephones and power to the land owner, which is a significant cost item.

The project is also being forced into an unfavourable position by one part owner of an estate where the powerhouse is to be situated. The partner of one of the heirs to this estate has refused to enter into an agreement that the other 6 heirs entered into. Instead, he demanded an exorbitant payment 10 times higher than the fair value accepted by the other owners. This is an untenable situation and the Project has investigated relocating the powerhouse to avoid this estate. However, following additional engineering studies, the only other possible location is not feasible as it is not located on bedrock. The new site for the powerhouse would require at least 35m of additional excavation to reach bedrock. While this would slightly increase the generation of the Project, the cost is in the order of \$10m additional investment. This is not an economical alternative. Of even more importance to the Project is the delay in receiving a new Environmental Permit (at least 6 months) and a new Hydraulic Works Permit (4 months) if forced to relocate the powerhouse. To ensure the project is not delayed the Owners may have to pay well over market rates to secure access.

Similar barriers are not faced by natural gas fired plants, for example, where total project area is a lot smaller and there is much more flexibility to optimally locate based on land issues.

Financing Risk:

The Project company, Hidroeléctrica La Higuera, is a 50% JV between PHL and SNPI. PHL has AU\$271m in net assets, and this project as a whole would represent 113% of the company's net assets. As such it is not possible to secure the project investment via corporate financing and the company's balance sheet. On this basis the company sold 50% of the project to SNPI in May 2004, to reduce its funding commitment. As SNPI has only been in operation for 3 years as a development fund it is not able to fund projects via balance sheet financing. As a result the project has had to seek non recourse project finance, where only the assets of the project secure the loan. Banks require 50-100% of revenue to be subject to sales contract to PPAs to guarantee funds for debt payments. Typically Debt Service Cover Ratios of 2 or higher are required where the lenders have some exposure to spot market risk (merchant risk). The blackouts in California highlighted this exposure of the power sector, and as result banks are very risk averse to any spot market exposure. In the first half of 2003, the Sponsors approached the commercial market for non recourse project financing. Given the Californian situation, the power and corporate crisis triggered by Argentinean devaluation, and Brazilian energy crisis, no commercial banks would make non recourse project financing available for longer than 6 years. They also included a requirement for contracting the sale of 100% of plant generation. As discussed above the project could not support supply risk of 100% contractual obligations, and hence had to seek alternative sources. In June 2003 the Project entered into a Mandate Letter with IFC to raise financing of the project. IFC is the World Bank's private investment bank, with a mandate to finance private companies and projects in under developed and developing countries. They typically only enter into projects and countries that Commercial Banks will not do on their own right, or at very unfavourable terms to the borrowers. On this



basis they are known in the market as ‘lenders of last resort’.³ As such the requirements they place on borrowers are much more onerous and not what is obtainable under commercial financing. They also do not underwrite the deal, so if enough banks cannot be enticed to join the syndication, the project remains unfunded. They are similarly limited to 25% of the project costs, so that commercial banks at the end of the day take up the largest part of the financing.

The Project has recently received provisional financing commitments from Syndicate banks through an IFC syndication process. The following is the probable financing composition for the Project:

SOURCE	\$US(000s)	%
<i>Debt</i>		
IFC A Loan	35,000	13.3%
Syndicated B Loan	115,000	43.7%
IFC C Loan	10,000	3.8%
Total Debt	160,000	60.8%
<i>Equity</i>		
PHL	51,500	19.6%
SNPI	51,500	19.6%
Total Equity	103,000	39.2%
Total Project Funding	263,000	100.0%

Table 4 Financing Plan

Spot Market Risk:

Any generation activity has an element of spot market risk, since prices are a result of market forces, in this case fuel costs and hydrology are the main price factors. As the SIC has a 60% hydroelectricity base variability of spot prices is exaggerated. During a normal or wet hydrological year spot prices are very low, severely impacting project revenues. As mentioned above the project activity does not have any storage allowing for flow regulation so project will also suffer periods of very high spot prices when there is a drought, and it has no ability to regulate flows. As discussed below there is contracting risk as a result of being required to have a power purchase agreement (PPA) in place for lenders. HLH were recently able to obtain a PPA from Chilectra, Chile’s largest distributor, by providing energy at a discount to the regulated node prices. While this is confidential information this discount is in the order of \$US600,000-1,000,000 p.a. in foregone revenue to the project. This type of discount is not market practice with incumbent generators. This discount was the lowest that Chilectra would accept for a new company entering the system, and is related to their perceived uncertainty of project completion and the lack of other assets the company, HLH, has to supply under this PPA. Under non-recourse project financing the banks’ only security on receiving principal and interest (PI) payments is contracted energy sales. The high levels of revenues from contracted sales required to meet the PI (generally 1.3 to 1.75 times greater than PI) increases the project exposure to spot market to supply the client. This exposes the project to spot market risk in the months of April-August when contract commitments can be higher than plant generation. If there is a drought in these periods the project would have to purchase off the spot market to fulfil its contract with the offtaker.

³ A. Hawser. A Matter of Principles. *GLOBAL FINANCE MAGAZINE*, January 2005



Market and Regulatory risk:

Despite being a highly developed, deregulated and transparent system, the Chilean Electricity sector is still subject to changes that provide considerable barriers to the project. The first example of this is the introduction of Article 99bis, following a severe drought in 1999. Generators were unable to supply under their contracts to distributors and claimed *force majeure*. To try and reduce the likelihood of blackouts in the future, the authorities removed drought as an allowable *force majeure* event, thus making the generators responsible for finding other ways of securing delivery. The result has been that generators

without a diverse portfolio of hydro and thermal generation assets have not been able to sign PPAs with distributors, who fall under Article 99bis jurisdiction. Since distributors are 80% of the market for sales, it severely limits the project to find financially attractive PPA to ensure financing.

The second example is current changes being made by CNE with respect to the period for which Firm Capacity is calculated. In October 2004, they changed the 5 hour period between 6 and 11 pm between the months of May and September to 8 hours. As the project has very limited ability for storage, currently only 5 hours due to geographical and environmental constraints, CDEC would downgrade the assigned firm capacity of the Plant. Firm capacity revenues for 5 hours represent 30% of project revenues, and changing to an 8 hour period would reduce firm capacity revenues by 37%.

Sub-step 3b. Show that the identified barriers would not prevent the implementation of at least one of the alternatives

Natural gas combined or open cycle plants are considered the least cost investment by CNE, the governmental Electricity Authority. On this basis CNE uses combined cycle gas plants as the default investment in planning system expansion, “Plan de Obras”. They can easily be shown to not suffer the barriers identified above.

Hydrology Risk:

Natural gas plants are not constrained by hydrological conditions. While there may be some fuel supply risk for these plants they typically can run on either natural gas or diesel, the latter of which is readily available.

Tunnelling Risk:

Technological risk, in particular tunnelling is not relevant to gas fired plants. Construction techniques and technology are well established, resulting in a lower investment cost.

Land Risk:

The physical requirements for land for a combined cycle natural gas plant are considerably less than the proposed project activity. Indeed these plants can typically be built almost anywhere provided there is the ability to transport and access natural gas. This project activity is specifically tied to the unique characteristics of the topography, utilising available water and optimising construction cost with gross head. As such the project is dependent on the willingness and reasonableness of landowners in the project area. The Owner has experienced lengthy and negative sentiment by some of the landowners in the area, actually being forced to change the layout of the project once already.

**Financing Risk:**

In Chile the electricity generation sector is entirely open to investment by any company. The type of financing used is obviously dependent on size and the corporate strategies each investor has. However, due to hydrology risk, large scale run of river hydroelectric projects would require the investor to have their own alternate thermal generation, or upstream regulation. As such, large scale projects in general would be restricted to companies with an installed base of thermal assets, or if not the latter, the ability to raise financing without high contracted power sale levels. A natural gas plant does not face the same funding hurdles.

A 350MW combined cycle plant will typically have an investment cost of \$US208m, compared with the 155 MW La Higuera plant of \$US225m. The project activity is also not the most cost effective investment when annualised costs of production are considered. Table 4 shows that the annualised cost of investment and generation of a natural gas plant over the net present generation during the life of the plant is lower for a natural gas plant than for the project activity⁴.

Summary	La Higuera Project Activity	Combined Cycle Gas
Investment Cost (\$m)	225.0	207.6
Annual Fixed Costs (\$m)	3	3.4
Annual generation (GWh)	823	2818
PV of Energy (GWh)	4,804.6	19,781.1
PV of Annual and investment costs (\$m)	-\$215,536.11	-\$742,006.54
Average cost of generation \$US/MWh	44.9	37.5

Table 5 - Average annualised cost of generation comparison

Spot Market Risk:

As discussed above natural gas plants are also able to use diesel, thus mitigating supply risks that run of river projects, like the project activity, face. On this basis this barrier is not as extreme for a natural gas plant as the project activity.

Market and Regulatory risk:

As mentioned the project is particularly susceptible to changes in regulation governing the authority's recognition of the plants firm power. Water, environmental and geographical restraints restrict the ability for the project to adapt. A natural gas plant does not face the same restrictions, and generally such flexibility has a lower capital cost for these types of projects.

On the basis of the above analysis it is obvious that the project activity faces unique, identifiable and significant barriers, not faced by its most likely alternative, a gas fired plant.

Step 4. Common Practice analysis

Sub-step 4a. Analyse other activities similar to the proposed project activity

Large Run of River hydroelectric projects that are not connected to reservoir facilities are not common

⁴ 11% discount rate is used for both technologies. Given the longer construction period, high risk geographical location and hydrological exposure for the Project Activity it would be appropriate to consider a higher discount rate for the Project Activity. For conservatism the same rate for both options has been used for this analysis.

within SIC. The following table lists the hydropower plants connected to the Chilean electricity grid constructed over the last 25 years. From the table below, the only plants of similar size are Rucúe and Alfalfal. Rucúe is downstream of the Laja reservoir, and is able to directly benefit from the storage facility of this very large dam. As such it has a very different hydrological risk to a run of river project that has no storage facility upstream.. The other similar plant is Alfalfal. This plant was built in 1991 (16 years ago) when the energy and financing situation where completely different and cannot be compared with the current situation. This will be discussed in the next section.

Name of Project	Installed Capacity	Year of completion	Associated with large reservoirs
Chacabuquito	25	2002	no
Mampil	49	2000	no
Peuchén	75	2000	no
Rucúe	170	1998	Laja
Puntilla	14.7	1997	no
Loma Alta	38	1997	Maule, Invernada
San Ignacio	37	1996	Colbún
Capullo	10.7	1995	no
Aconcagua	72.9	1993	no
Curillínque	85	1993	Maule, Invernada
Alfalfal	160	1991	no

Table 6 - List of run of river plants constructed over the last 25 years. **Source:** SIC

Sub-step 4b. Discuss any similar options that are occurring:

Large Run of River hydroelectric projects that are not connected to reservoir facilities are rare within the project boundary, SIC. La Higuera will be the second biggest run of river project ever built by installed capacity in Chile⁵. In 1991 the 160 MW Alfalfal plant was built on the Colorado River, some 80km north of the La Higuera project. This project is similar in many ways to La Higuera project, however, the economics of the Alfalfal project were significantly better for two main reasons. Firstly, in 1991 there was no gas available, and as such node prices for energy were significantly higher than current prices. The second factor is that the owners, GENER, also have coal fired thermal plants, allowing them to mitigate the high seasonal generation of low winter generation and high summer snow melt generation. Following the introduction of imported natural gas from Argentina in 1996 there was significant change in the electricity sector's structure, with high growth in the lower capital cost and shorter payback period combined cycle natural gas plants as shown in Figure 3. This resulted in a significant reduction in node prices. This has increased the barriers to hydroelectricity and other renewable energy technologies, which are typically much higher capital cost investments.

Step 5. Impact of CDM registration

This section clearly explains how the approval and registration of the project as a CDM activity, and the attendant benefits and incentives derived from the project activity, will alleviate the barriers showed in Step 3 and thus enable the project to be undertaken. The benefits and incentives brought about by the CDM to the shareholders of the proposed project activity are as follows:

- The financial benefit from the revenue obtained by selling the CO2 emissions reductions has been one of the key issues that encourage shareholders in the Project company to invest in the

⁵ This does not include run of river plants associated with large reservoirs, which provide annual regulation, and as such should not be considered as run of river projects.

proposed project activity. The CDM has been considered from an early stage and it is an integral part of the financial package of the proposed project activity.

- Attracting new players who bring the capacity to implement a new technology. In this case, Pacific Hydro Ltd., an Australian company, and SNPI, a Norwegian company, are both entering a new market. The value of the CDM registration of investments is an important factor in attracting such world leading companies to Chile, bringing capacity and technology.

Therefore, it has been clearly demonstrated how the approval and registration of the project as a CDM activity, and the attendant benefits and incentives derived from the project activity, will alleviate the barriers showed in Step 3 and thus enable the project to be undertaken.

So following all steps of the additionality test, it can be clearly demonstrated that the proposed CDM project activity is not the baseline scenario.

B.4. Description of how the definition of the project boundary related to the baseline methodology selected is applied to the project activity:

>>

Chile has four different grids and there are no interconnections between them. Therefore each grid defines the geographical and system boundaries for proposed projects located within it (see map below). The Central Interconnected Grid (SIC), to which La Higuera project is connected, comprises the regions 3 to 10 and accounts 64 percent of the total capacity. The project boundary of La Higuera project is the hydroelectric plant and all power plants connected physically to the SIC grid. The SIC grid electricity system's boundary is limited to the spatial extent of the power plants connected to the SIC grid, that can be dispatched without significant constraints. The following diagrams illustrate the location and spatial extent of the SIC grid

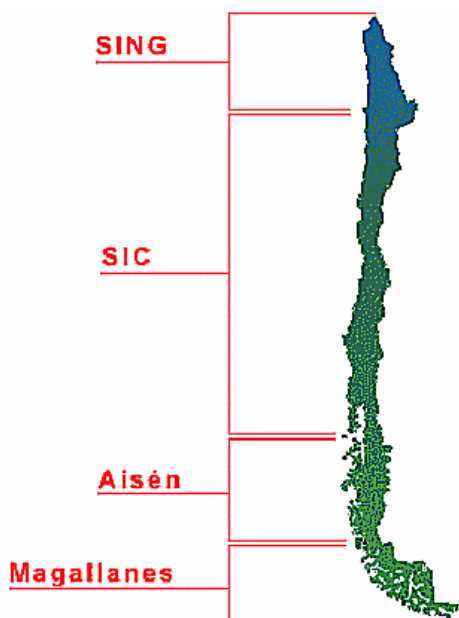


Figure 5 - Chilean Grid Systems

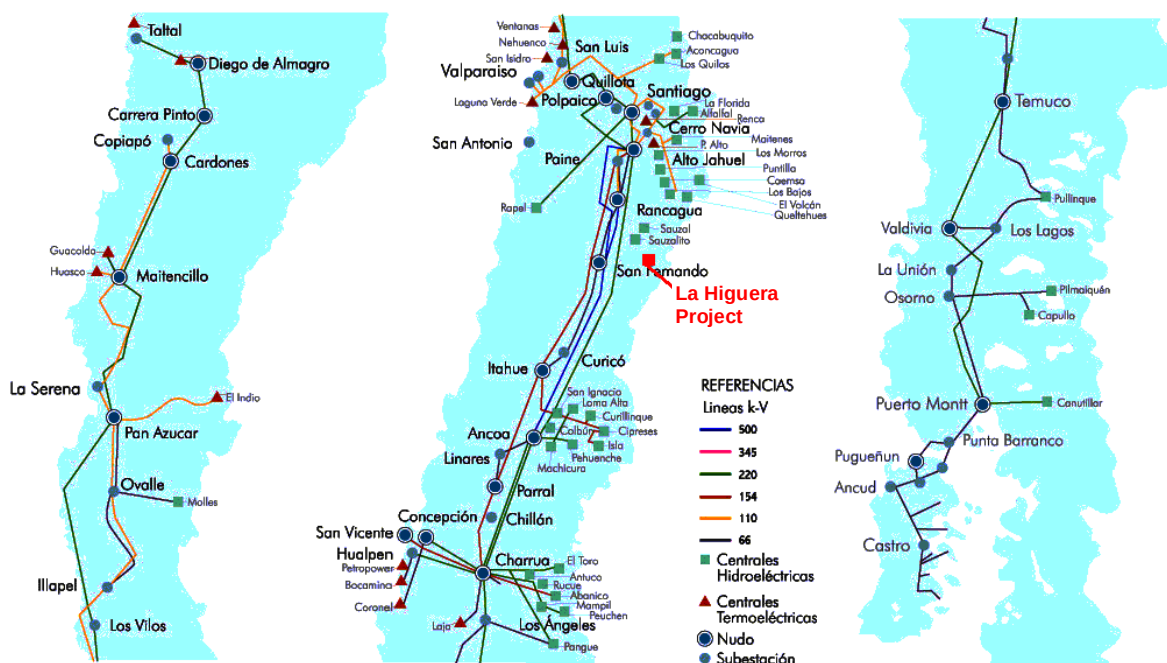


Figure 6 - The SIC System.

B.5. Detailed baseline information, including the date of completion of the baseline study and the name of person (s)/entity(ies) determining the baseline:

>>>

The baseline study was completed on 15/06/2005. The entity determining the baseline as the Carbon Advisor is EcoSecurities Ltd., UK. The baseline information (baseline emission factor for the SIC electricity system) was revised on June 2011 with Poch Ambiental support.

Contact information for EcoSecurities:

Pedro Moura Costa
EcoSecurities Ltd.
21 Beaumont Street
Oxford OX1 2NH
Telephone (44) 1865 202635
Fax: (44) 1865 251438
www.ecosecurities.com

Contact information for Poch Ambiental S.A:

Renato Sánchez 3838 – Las Condes
Santiago, Chile.
Telephone: (56)-(2) 2070154
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Consultants:

María Luz Farah Gonzales (marialuz.farah@poch.cl)
Mariela Ramos Marín (mariela.ramos@poch.cl).

**SECTION C. Duration of the project activity / Crediting period****C.1 Duration of the project activity:****C.1.1. Starting date of the project activity:**

>>

Project development commenced in January 2002 when Pacific Hydro purchased on Option to Purchase the Water rights for the Tinguiririca Valley. Project Construction is expected to commence in the first quarter of 2006 and the project is to be connected to the grid and generating by January 2008.

C.1.2. Expected operational lifetime of the project activity:

>>

25 years, 0 Months

C.2 Choice of the crediting period and related information:

>>

The project will use a renewable crediting period

C.2.1. Renewable crediting period**C.2.1.1. Starting date of the first crediting period:**

>>

Expected 01/01/2008 (if registration has been completed)

C.2.1.2. Length of the first crediting period:

>>

7 years

C.2.2. Fixed crediting period:**C.2.2.1. Starting date:**

>>

Not applicable

C.2.2.2. Length:

>>

Not applicable

**SECTION D. Application of a monitoring methodology and plan****D.1. Name and reference of approved monitoring methodology applied to the project activity:**

>>

Approved consolidated monitoring methodology ACM0002: “Consolidated monitoring methodology for grid-connected electricity generation from renewable sources.” (Version 4)

D.2. Justification of the choice of the methodology and why it is applicable to the Project activity:

>>

The project activity is a grid-connected run of river hydropower project, where the grid’s geography and system boundaries are explicit and characteristics are available through CNE and CDEC-SIC. On this basis the conditions for applying ACM0002 are met.



D.2.1. Option 1: Monitoring of the emissions in the project scenario and the baseline scenario:

D.2.1.1. Data to be collected in order to monitor emissions from the project activity, and how this data will be archived:

ID number (Please use numbers to ease cross referencing to D.3)	Data variable	Source of data	Data unit	Measured (m), calculated (c) or estimated (e)	Recording frequency	Proportion of data to be monitored	How will the data be archived? (electronic/paper)	Comment															
15. Hours of operation	Hours of operation of each generator i	Obtained from Hour-meter of each gen-set	Hr	m	monthly	100%	electronic																
16. Fuel Consumption (FCi)	Fuel consumption of each generator i	Calculated based on hours of opration and default diesel consumption at full load (according to the manufacturer)	tons	c	monthly	100%	electronic	This value is obtained by multiplying the hours of operation with the diesel consumption at full load (manufacturer information)															
17. Net calorific value (NCV _{gen-set})	Net calorific value of diesel	IPCC, higher value at 95% confidence level	TJ/Gg	e	yearly	100%	electronic	IPCC, higher value at 95% confidence level: 43.3 TJ/Gg															
18. Emission factor of fuel (EF _{gen-set})	CO2 emission factor per unit of energy	IPCC, higher value at 95% confidence level	tCO ₂ /TJ	e	yearly	100%	electronic	IPCC, higher value at 95% confidence level: 74.8 tCO2/TJ															
19. Spetic diesel consumption at full load	Diesel consumption at full load of each generator i	Manufacturer information	t diesel/hr	e	yearly	100%	electronic	The specific fuel consumption for the gen-sets are:<table><tr><th>Location</th><th>i</th><th>ton/hr</th></tr><tr><td>La Higuera</td><td>1</td><td>0.1139</td></tr><tr><td>Azufre Intake</td><td>2</td><td>0.0140</td></tr><tr><td>Tinguiririca</td><td>3</td><td>0.0119</td></tr><tr><td>Los Helados</td><td>4</td><td>0.0119</td></tr></table>	Location	i	ton/hr	La Higuera	1	0.1139	Azufre Intake	2	0.0140	Tinguiririca	3	0.0119	Los Helados	4	0.0119
Location	i	ton/hr																					
La Higuera	1	0.1139																					
Azufre Intake	2	0.0140																					
Tinguiririca	3	0.0119																					
Los Helados	4	0.0119																					

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**D.2.1.2. Description of formulae used to estimate project emissions (for each gas, source, formulae/algorithm, emissions units of CO₂ equ.):**

>>

ACM0002 does not require calculation of project emissions in case of hydro power projects. However, for reasons of conservativeness, project emission from gen-sets that will be operated in case of grid failures are taken into consideration as follows:

Emissions of each gen-set is obtained based on metering of the operating hours through the calculation of diesel consumption, the calorific value and the emission factor of the diesel. The default values of NCV and emission factor of the diesel will be derived from the IPCC, higher value at 95% confidence level.

$$PE = \sum FC_i * NCV * EF$$

Where:

FC_i: Diesel consumption of gen-set i

NCV: Net Calorific Value of diesel (default value of 0.0433 TJ/t IPCC, higher value at 95% confidence level)

EF: Diesel emission factor, (default value of 74.8 tCO₂/TJ of IPCC, i.e. higher value at 95% confidence level)

Where the fuel consumption is calculated as

$$FC_i = \text{Hours of operation of gen-set } i * \text{specific diesel consumption at full load of gen-set } i$$

D.2.1.3. Relevant data necessary for determining the baseline of anthropogenic emissions by sources of GHGs within the project boundary and how such data will be collected and archived⁶:

ID number	Data variable	Source of data	Data unit	Measured (m), calculated (c) or estimated (e)	Recording frequency	Proportion of data to be monitored	How will the data be archived? (electronic/ paper)	Comment
1. EG _y	Electricity supplied to the grid by the project	Metering system readings	MWh	Directly measured meter results (m) used to determine EG _y	Hourly measurement and monthly recording	100%	electronic	Electricity supplied to the SIC grid by the project activity determined based on equation 7 as explained under D.2.4 (Measurement of EG _y). Meters involved in obtaining the parameter are described below (1.1 to 1.6).

⁶ Archived data will be kept during the crediting period and two years after.

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1.1 M5	Electricity supplied to the grid by the project	Metering system readings	MWh	Directly measured meter results (m) used to determine EG_v	Hourly measurement and monthly recording	100%	electronic	M5, meter that records energy supplied to the grid from Tinguiririca substation
1.2 M6	Electricity supplied to the grid by the project	Metering system readings	MWh	Directly measured meter results (m) used to determine EG_v	Hourly measurement and monthly recording	100%	electronic	M6, meter that records energy supplied to the grid from Tinguiririca substation
1.3 M3	Electricity supplied by the La Higuera project	Metering system readings	MWh	Directly measured meter results (m)	Hourly measurement and monthly recording	100%	electronic	M3, meter that records energy from La Higuera hydropower in La Higuera substation
1.4 M4	Electricity supplied by the La Higuera project	Metering system readings	MWh	Directly measured meter results (m)	Hourly measurement and monthly recording	100%	electronic	M4, meter that records energy from La Higuera hydropower in La Higuera substation
1.5 M7	Electricity supplied from the La Confluencia substation	Metering system readings	MWh	Directly measured meter results (m)	Hourly measurement and monthly recording	100%	electronic	M7, meter that records energy from La Confluencia substation in La Higuera substation
1.6 M8	Electricity supplied from the La Confluencia substation	Metering system readings	MWh	Directly measured meter results (m)	Hourly measurement and monthly recording	100%	electronic	M8, meter that records energy from La Confluencia substation in La Higuera substation
1.7 M1	Electricity generated at the La Higueara project	Metering system readings	MWh	Directly measured meter results (m)	Hourly measurement and monthly recording	100%	electronic	M1, meter that records energy generated at the La Higuera powerhouse
1.8 M2	Electricity generated at the La Higueara project	Metering system readings	MWh	Directly measured meter results (m)	Hourly measurement and monthly recording	100%	electronic	M2, meter that records energy generated at the La Higuera powerhouse
2. EF_y	CO ₂ emission factor of the grid	Calculated ex-post using CDEC-SIC operation statistics, data published by CNE and IPCC	tCO ₂ /MWh	c	yearly	100%	electronic	Calculated as a weighted sum of the OM and the BM emission factor

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		default values						
3. $EF_{OM,y}$	CO ₂ Operating Margin emission factor of the grid (Simple Adjusted OM)	Calculated ex-post using CDEC-SIC operation statistics, data published by CNE and IPCC default values.	tCO ₂ /MWh	c	yearly	100%	electronic	Simple Adjusted OM updated based on ex post monitoring for the year in which project generation occurs
4. $EF_{BM,y}$	CO ₂ Build Margin Emission factor of the grid	Calculated ex-ante using CDEC-SIC data, published by CNE and IPCC default values.	tCO ₂ /MWh	c	yearly	100%	electronic	Calculated as $[\sum_i F_{i,y} * COEF_i] / [\sum_m GEN_{m,y}]$ over recently built power plants defined in the baseline methodology at the beginning of each crediting period. The calculation of the ex-ante value will be performed at the beginning of each period.
5. $F_{i,j/k,y}$	Amount of each fossil fuel consumed by each power source / plant j or k	CDEC-SIC operation statistics	Mass or volume	m	yearly	100%	electronic	The annual fuel consumption $F_{i,j/k,y}$ of each power unit is sourced from the yearbook of the CDEC, available both per power plant and per type of fuel
6. $COEF_i$	CO ₂ emission coefficient of each fuel type i	$COEF_i$ is obtained as per equation (2) of section D.2.1.4	tCO ₂ / mass or volume unit	c	yearly	100%	electronic	Plant or country specific values to calculate COEF are preferred to IPCC default values.
7. $GEN_{j/k,y}$	Electricity generation of each power source / plant j or k	CDEC-SIC operation statistics	MWh/a	m	yearly	100%	electronic	Obtained from the dispatch center
8. Plant name OM	Identification of power source / plant for the OM	CDEC-SIC operation statistics	Text	e	yearly	100% of set of plants	electronic	Identification of plants (j, k) to calculate Simple Adjusted Operating Margin emission factors
9. Plant name BM	Identification of power source / plant for the BM	CDEC-SIC operation statistics	Text	e	yearly	100% of set of plants	electronic	Identification of plants (m) to calculate Build Margin emission factors
10. λ_y	Fraction of time during which low-cost/ must-run sources	CDEC-SIC operation statistics	Number	c	yearly	100%	electronic	Factor accounting for number of hours per year during which low-cost/ must-run sources are on the margin

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	are on the margin							
11a. $GEN_{j/k/l,y}$ IMPORTS	Electricity imports to the project electricity system	CNE ⁷	kWh	c	yearly	100%	electronic	Obtained from the latest local statistics (CNE). If local statistics are not available, IEA statistics are used to determine imports.
11b. $COEF_{ij,y}$ IMPORTS	CO ₂ emission coefficient of fuels used in connected electricity systems (if imports occur it will be calculated as per equation (2) of section D.2.1.4))	Obtained from the latest local statistics (CNE).	tCO ₂ / mass or volume unit	c	yearly	100%	electronic	Obtained from the latest local statistics (CNE). If local statistics are not available, IPCC default values are used to calculate.
12. NCV_i	Net calorific value of fuel <i>i</i>	CNE Node Price Report and/or CNE National Energy Balance	Energy content per mass or volume unit of a fuel <i>i</i>	Calculated (c) from the gross calorific value (GCV) of each fuel type obtained from National Energy Balance and then converted into NCV	yearly	100%	electronic	Conversion from GCV reported to NCV for each type of fuel <i>i</i> is made by applying guidance from 2006 IPCC Guidelines for National Greenhouse Inventories, as follows: $NCV_i = GCV_i \cdot CF_i$ where GCV _{<i>i</i>} represents gross calorific value for fuel <i>i</i> and CF _{<i>i</i>} the Conversion Factor from gross to net calorific value. (CF for oils and solids = 0.95 and for gas = 0.90).
13. $OXID_i$	Oxidation factor of the fuel	2006 IPCC Guidelines for National Greenhouse Gas Inventories, Table 1.4, Chapter 1, Vol. 2	%	e	yearly	100%	electronic	1996 Revised IPCC Guidelines for default values applied for <i>ex-ante</i> BM calculation. For <i>ex-post</i> calculation default carbon oxidation factor as per 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Table 1.4, Chapter 1, Vol. 2 will be used
14. $EF_{CO_2,i}$	CO ₂ emission factor per unit of energy of the	2006 IPCC Guidelines for National	tCO ₂ / unit of energy	e	yearly	100%	electronic	Where available, local values should be used. If no such values are available, country-specific values (see e.g. IPCC

⁷ Comision Nacional de Energia, Chile



	fuel <i>i</i>	Greenhouse Gas Inventories, Table 1.4, Chapter 1, Vol. 2						Good Practice Guidance) are preferable to IPCC world-wide default values.
--	---------------	--	--	--	--	--	--	---

D.2.1.4. Description of formulae used to estimate baseline emissions (for each gas, source, formulae/algorithm, emissions units of CO₂ equ.):

>>

For project activities that do not modify or retrofit an existing electricity generation facility, the baseline scenario is the following:

Electricity delivered to the grid by the project would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described below.

The baseline emission factor (EF_y) is calculated as a combined margin (CM), consisting of the combination of operating margin (OM) and build margin (BM) factors according to the following three steps. Calculations for this combined margin must be based on data from an official source (where available) and made publicly available.

STEP 1. Calculate the Operating Margin emission factor(s) ($EF_{OM,y}$) based on one of the four following methods:

- (a) Simple OM, or
- (b) Simple adjusted OM, or
- (c) Dispatch Data Analysis OM, or
- (d) Average OM.

Dispatch Data Analysis should be the first methodological choice. Since the fuel consumption - Parameter 5 $F_{i,y}$ as per methodology ACM0002 version 4 - is not available measured on an hourly basis, but only the yearly specific fuel consumption per plant is reported⁸, the Dispatch Data Analysis is not applicable and alternatively Simple Adjusted OM may be used. In absence of the Dispatch Data Analysis the option Simply Adjusted OM is deemed the most accurate method to calculate the OM. In this case, yearly fuel consumption per each grid connected plant is available in a CDEC-SIC operational statistics yearbook. The Simple Adjusted OM has been chosen and is applied for the project activity as follows.

(b) *Simple Adjusted OM*. This emission factor ($EF_{OM, simple\ adjusted,y}$) is a variation on the Simple OM method, where the power sources (including imports) are separated in low-cost/must-run power sources (*k*) and other power sources (*j*):

⁸ The information provided by the official source CNE is the specific fuel consumption on an annual basis only; hourly fuel consumption is not reported and monitored. This template shall not be altered. It shall be completed without modifying/adding headings or logo, format or font.



$$EF_{OM, simple_adjusted, y} = (1 - \lambda_y) \cdot \frac{\sum_{i,j} F_{i,j,y} \cdot COEF_{i,j}}{\sum_j GEN_{j,y}} + \lambda_y \cdot \frac{\sum_{i,k} F_{i,k,y} \cdot COEF_{i,k}}{\sum_k GEN_{k,y}} \quad (1)$$

where $F_{i,j,y}$, $COEF_{i,j}$ and GEN_j are analogous to the variables described for the Simple OM method for plants j :

$F_{i,j,y}$ is the amount of fuel i (in a mass or volume unit) consumed by relevant power sources j in year(s) y ,
 j refers to the power sources delivering electricity to the grid, not including low-operating cost and must-run power plants, and including imports⁹ to the grid
 $COEF_{i,j,y}$ is the CO₂ emission coefficient of fuel i (tCO₂ / mass or volume unit of the fuel), taking into account the carbon content of the fuels used by relevant power sources j and the per cent oxidation of the fuel in year(s) y
 $GEN_{j,y}$ is the electricity (MWh) delivered to the grid by source j .

The variables $F_{i,k,y}$, $COEF_{i,k}$ and GEN_k are accordingly to be used for plants k .

The CO₂ emission coefficient $COEF_i$ is obtained as

$$COEF_i = NCV_i \cdot EF_{CO_2,i} \cdot OXID_i \quad (2)$$

where:

NCV_i is the net calorific value (energy content) per mass or volume unit of a fuel i
 $OXID_i$ is the oxidation factor of the fuel $EF_{CO_2,i}$ is the CO₂ emission factor per unit of energy of the fuel i .

Where available, local values of NCV_i and $EF_{CO_2,i}$ should be used. If no such values are available, country-specific values (see e.g. IPCC Good Practice Guidance) are preferable to IPCC world-wide default values.

The years(s) y can reflect either of the two vintages noted for Simple OM¹⁰, of which the *ex-post* data vintage has been chosen, and λ_y is defined as follows

⁹ As described in ACM0002 version 4, an import from a connected electricity system should be considered as one power source j .

¹⁰ The Simple OM emission factor can be calculated using either of the two following data vintages for years(s) y : A 3-year average, based on the most recent statistics available at the time of PDD submission, or the year in which project generation occurs, if $EF_{OM,y}$ is updated based on ex post monitoring. This template shall not be altered. It shall be completed without modifying/adding headings or logo, format or font.



$$\lambda_y = \frac{\text{number of hours per year for which low – cost / must – run sources are on margin}}{8760 \text{ hours per year}} \quad (3)$$

where lambda (λ_y) should be calculated as follows (see figure below):

- Step i) Plot a Load Duration Curve. Collect chronological load data (typically in MW) for each hour of a year, and sort load data from highest to lowest MW level. Plot MW against 8760 hours in the year, in descending order.
- Step ii) Organize Data by Generating Sources. Collect data for, and calculate total annual generation (in MWh) from low-cost/must-run resources (i.e. $\sum_k GEN_{k,y}$).
- Step iii) Fill Load Duration Curve. Plot a horizontal line across load duration curve such that the area under the curve (MW times hours) equals the total generation (in MWh) from low-cost/must-run resources (i.e. $\sum_k GEN_{k,y}$).
- Step iv) Determine the "Number of hours per year for which low-cost/must-run sources are on the margin". First, locate the intersection of the horizontal line plotted in step (iii) and the load duration curve plotted in step (i). The number of hours (out of the total of 8760 hours) to the right of the intersection is the number of hours for which low-cost/must-run sources are on the margin. If the lines do not intersect, then one may conclude that lowcost/must run sources do not appear on the margin and λ_y is equal to zero. Lambda (λ_y) is the calculated number of hours divided by 8760.

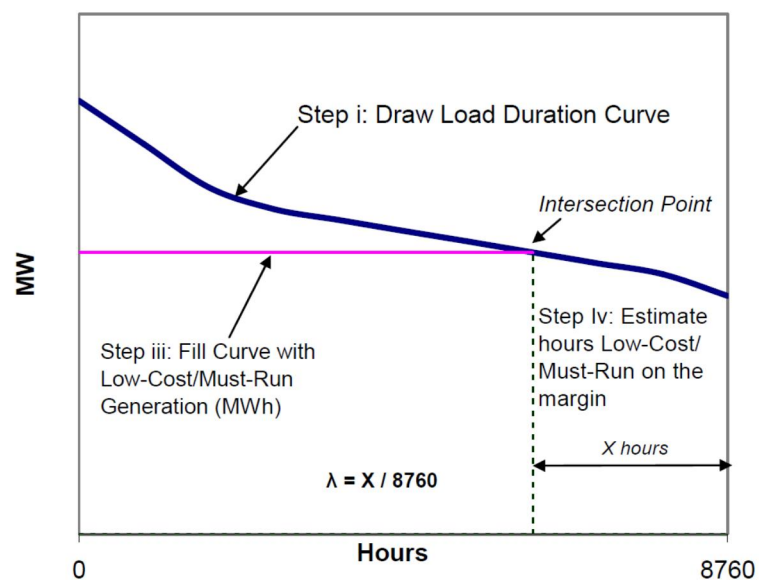


Figure 7: Illustration of Lambda Calculation for Simple Adjusted OM Method

Note: Step (ii) is not shown in the figure; it deals with organizing data by source.

STEP 2. Calculate the Build Margin emission factor ($EF_{BM,y}$)

The calculation of the Build Margin emission factor is provided in Annex 3; this value is fixed *ex-ante* for the first crediting period.

STEP 3. Calculate the baseline emission factor EF_y

The baseline emission factor EF_y is calculated as the weighted average of the Operating Margin emission factor ($EF_{OM,y}$) and the Build Margin emission factor ($EF_{BM,y}$) as outlined in the registered PDD:

$$EF_y = w_{OM} \cdot EF_{OM,y} + w_{BM} \cdot EF_{BM,y} \quad (4)$$

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where the weights w_{OM} and w_{BM} , by default, are 50% (i.e., $w_{OM} = w_{BM} = 0.5$), and $EF_{OM,y}$ and $EF_{BM,y}$ are calculated as described in Steps 1 and 2 above and are expressed in tCO₂/MWh.

D. 2.2. Option 2: Direct monitoring of emission reductions from the project activity (values should be consistent with those in section E).

>>

Not applicable

D.2.2.1. Data to be collected in order to monitor emissions from the project activity, and how this data will be archived:

ID number (Please use numbers to ease cross referencing to D.3)	Data variable	Source of data	Data unit	Measured (m), calculated (c) or estimated (e)	Recording frequency	Proportion of data to be monitored	How will the data be archived? (electronic/paper)	Comment

>>

Not applicable

D.2.2.2. Description of formulae used to calculate project emissions (for each gas, source, formulae/algorithm, emissions units of CO₂ equ.):

>>

Not applicable

D.2.3. Treatment of leakage in the monitoring plan**D.2.3.1. If applicable, please describe the data and information that will be collected in order to monitor leakage effects of the project activity**

ID number (Please use numbers to ease cross referencing to D.3)	Data variable	Source of data	Data unit	Measured (m), calculated (c) or estimated (e)	Recording frequency	Proportion of data to be monitored	How will the data be archived? (electronic/paper)	Comment

>>

Not applicable

**D.2.3.2. Description of formulae used to estimate leakage (for each gas, source, formulae/algorithm, emissions units of CO₂ equ.):**

>>

Not applicable

D.2.4. Description of formulae used to estimate emission reductions for the project activity (for each gas, source, formulae/algorithm, emissions units of CO₂ equ.)

>>

Emission reduction estimation

The project activity mainly reduces carbon dioxide through substitution of grid electricity generation with fossil fuel fired power plants by renewable electricity. The emission reduction ER_y by the project activity during a given year y is the difference between baseline emissions (BE_y), project emissions ($PE_y = 0$) and emissions due to leakage ($L_y = 0$), as follows:

$$ER_y = BE_y \quad (5)$$

where the baseline emissions (BE_y in tCO₂) are the product of the baseline emissions factor (EF_y in tCO₂/MWh) calculated in Step 3 under D.2.1.4, times the electricity supplied by the project activity to the grid (EG_y in MWh), as follows:

$$BE_y = EG_y \cdot EF_y \quad (6)$$

For the project activity alike most renewable energy project activities, $PE_y = 0$.

Measurement of EG_y

The electricity produced by La Higuera hydro power project is transmitted to La Higuera Substation, which is located next to the power plant. This substation also receives electricity from La Confluencia hydro power project. La Higuera Substation is connected to the grid through the Tinguiririca Substation located near San Fernando. There is a double circuit transmission line (of approximately 38 km in length) between La Higuera substation and the Tinguiririca substation (SIC's injection substation).

As shown in Figure 8, the system has eight electricity meters relevant for the CDM monitoring, located at different stages. First, an electricity meter is placed at the terminals of each generator of La Higuera hydro power plant (meters M1 and M2). Second, two electricity meters (M3 and M4) are located in La Higuera Substation measuring electricity generated by La Higuera hydro power project, and another two electricity meters (M7 and M8) measure electricity from La Confluencia power plant. Finally, a set of two meters are placed in the Tinguiririca Substation (M5 and M6) in order to measure the energy generated.

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by La Higuera hydro power project plus La Confluencia hydro power project. As the electricity produced by La Higuera hydro power project is injected to the grid through the Tinguiririca Substation, main meters are considered to be M5 and M6. In addition to these eight meters, two meters situated at La Confluencia hydro power plant (meters M1_{HLC} and M2_{HLC} in Figure 2) are relevant to the Emergency Procedures.

A diagram of the location of these meters is presented in the following figure:

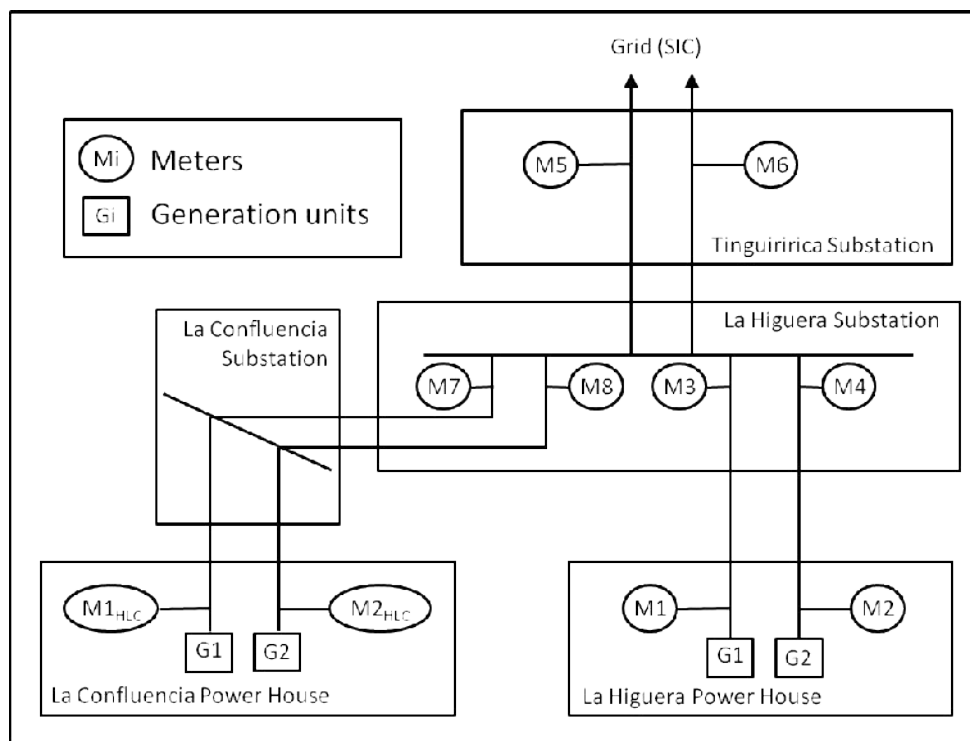


Figure 8: Diagram of monitoring points



The amount of electricity supplied to the grid by La Higuera hydro power project is measured along with the electricity supplied by La Confluencia hydro power project at the injection point located at the Tinguiririca substation owned by Transelec¹¹ (meters M5 and M6). In order to determine the amount of electricity supplied to the grid by La Higuera hydro power project at the injection point, the electricity injected by La Confluencia hydro power project is deducted from the measurements of meters M5 and M6.

The meters located at Tinguiririca substation (M5 and M6) and the meters located at La Higuera substation (M3, M4, M7 and M8) are bi-directional meters. This means that for every pulse the raw data from each meter contains a value associated to electricity injected and electricity retired. The net electricity value for each meter is the difference between the raw data of electricity injected and electricity retired. Then, using the net electricity values from each meter, the net electricity injected by La Higuera hydro power project at Tinguiririca substation is calculated by the Commercial Analyst using the following equation:

$$\text{Energy for invoicing HLH} = \text{EG}_y = (M5 + M6) \cdot \frac{(M3 + M4)}{(M3 + M4) + (M7 + M8)} \quad (7)$$

Where,

M_i Amount of electricity generation measured at electricity meter number i (according to Figure 1).

EG_y Amount of net electricity generated by La Higuera hydro power project.

This calculation assumes the proportional distribution of transmission losses in the Transmission line from La Higuera substation to Tinguiririca substation in function of the hourly energy injected by each power plant to La Higuera substation.

Data collection, recording, calculation

Electricity generated by La Higuera hydro power project and supplied to the grid is measured in the injection point at Tinguiririca Substation (meters M5 and M6). As mentioned above, meters located at this point are owned by Transelec.

A Field Technician from La Higuera hydro power project collects the data (electricity measurements every 15 minutes recorded in the memory of the meters) from the meters located in the Tinguiririca substation once a month, and delivers the data to the Operator of the power plant. The Operator is responsible for verifying the data collected is accurate and complete for every day of the month, in order to detect and inform any failure in the meters that needs to be repaired. Afterwards, the processed data is sent by the Operator to the Controller to cross check it with the meters of La Higuera substation. After his approval, the data is uploaded in the server.

¹¹



The Controller sends the processed data to the Commercial Analyst. Equation 7 above and Equation 9 below are applied by the Commercial Analyst in order to calculate the electricity generated by La Higuera hydro power project by deducting electricity from La Confluencia hydro power project.

Once the data is ready, the Commercial Analyst sends the information on a monthly basis to the CDEC-SIC (Economic Load Dispatch Centre of the SIC) for invoicing purposes and to the CDM Coordinator for emissions reduction calculations.

In all meters (M1 to M8) the measurements every 15 minutes are registered in an hourly basis with a daily recording frequency. Accuracy class of all meters (M1 to M8) is 0.2. The same measurement and recording frequency as well as the accuracy apply for the meters at La Confluencia hydro power plant (M1_{HI} and M2_{HLC}).

Data Quality Control

The procedure “Validation of meters’ data and monthly energy assignment for invoicing to the La Higuera hydro power project and La Confluencia hydro power project” ensures quality of the data measured at the grid injection point.

As a quality control procedure of the monitored data in the meters of Tinguiririca Substation (M5 and M6 in the Figure 2, which represents the electricity injection point to the grid where electricity from La Higuera hydro power project and La Confluencia hydro power project are provided), the monitored data is cross checked against the total of the data measured at the meters installed at the La Higuera Substation (M3+M4+M7+M8). In case of any inconsistency detected in the monitored data, wrong data or missing data will be completed as described in the same procedure.

Emergency Procedures

Failures of electricity metering at Tinguiririca substation:

In case of any failure on the data recording of Tinguiririca substation electricity meters (owned by Transelec) the generation data of the project at the injection point should be obtained by using the electricity measured at La Higuera substation with meters M3 + M4 (as in Figure 2) and deducting the typical average value of transmission losses in the La Higuera substation - Tinguiririca substation. This typical average value for transmission losses will be obtained based on previous records of the measurements at Tinguiririca substation and La Higuera substation, as is described in the procedure “Validation of meters’ data and monthly energy assignment for invoicing to the La Higuera hydro power project and La Confluencia hydro power project”.

Failures of electricity metering at La Higuera substation:

In case of metering failures of all the meters at La Higuera substation (M3, M4, M7 and M8) from Figure 2, while La Confluencia hydro power plant is operating, the following formula should be used in order to estimate electricity produced by La Higuera hydro power project:

$$EG_y = (M1+M2) - [(M1 + M2) + (M1_{HLC}+M2_{HLC}) - (M5 + M6)] / 2 \quad (8)$$

Where,

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$M5 + M6 =$ Represents the total amount of electricity measured in the electricity meter located at Tinguiririca substation that has been supplied by La Higuera and La Confluencia hydro power projects.
 $M1 + M2 =$ Represents the total amount of electricity measured at the generator terminals of La Higuera hydro power plant.
 $M1_{HLC} + M2_{HLC} =$ Represents the total amount of electricity measured at the generator terminals of La Confluencia hydro power plant.

In case of metering failures at one of the meters M3, M4, M7 or M8, or one of the pairs of meters M3 and M4 or M7 and M8, from Figure 2 located at La Higuera substation, the following formula should be used in order to estimate missing data from failed meters:

$$(M5 + M6) = [(M3 + M4) + (M7 + M8)] \cdot (1 - TL \text{ from HLH to TinSS}) \quad (9)$$

Where,

$M5 + M6 =$ Represents the total amount of electricity measured in the electricity meter located at Tinguiririca substation that has been supplied by La Higuera and La Confluencia hydro power projects.
 $M3 + M4 =$ Represents the electricity provided by La Higuera hydro power project and measured in the electricity meters located at La Higuera substation.
 $M7 + M8 =$ Represents the electricity provided by La Confluencia hydro power project and measured in the electricity meters located at La Higuera substation.
 $TL \text{ from HLH to TinSS} =$ Transmission losses from La Higuera substation to Tinguiririca substation.

The losses are calculated as the total generation at the La Higuera substation less the total injection at Tinguiririca substation, as follows.

$$TL = ((M7 + M8) + (M3 + M4)) - (M5 + M6) \quad (10)$$

Metering failures in all the main electricity meters outside the ownership control of the project participant (M3-M8):

Under this scenario, data should be calculated by using data from the electricity meters M1 and M2, located at the terminals of each unit of La Higuera hydro power project as needed. One meter is located at the terminals of each generation unit of the power plant connected to the La Higuera substation.

Transmission and other losses related to each unit should be deducted in order to reconstruct missing data due to the failure of electricity meters located at Tinguiririca or La Higuera substation according to the calculation methodology described in the procedure “Validation of meters’ data and monthly Energy assignment for invoicing to the La Higuera hydro power project and La Confluencia hydro power project”.

Any issues related to the Emergency Procedures such as meters failure will be included in the Monitoring Report of the corresponding monitoring period.

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In case of any difference between the energy generation data obtained from the meters and the one used for billing purposes by the CDEC, the more conservative will be used.

Verification procedures

The accuracy of all the electricity meters involved in the project activity (as shown in Figure 2: meters from Tinguiririca substation, from La Higuera substation and meters connected to the generation units terminals at the powerhouse), are verified every 2 years. The accuracy class of all meters is 0.2.

The verification and maintenance of the equipment is carried out by a reputed external entity (like Cam, Tecnet, or other companies) according to national standards. The verification of the meters which is performed with a 2 years frequency by an external entity consist in comparing the electricity measurement by the meter against a standard meter and if the meter presents an error on measurements results, then it is repaired or replaced if necessary. The external entity issues a certificate of the meter verification. These meters do not need to be calibrated periodically, they were calibrated by the manufacturer and are periodically verified, thus if the equipment has any measurement problem, it will be repaired or replaced.

Any failure of the monitoring equipment will be reported to the Lead of Generation, to coordinate the repair of the equipment.

D.3. Quality control (QC) and quality assurance (QA) procedures are being undertaken for data monitored		
Data (Indicate table and ID number e.g. 3.-1.; 3.2.)	Uncertainty level of data (High/Medium/Low)	Explain QA/QC procedures planned for these data, or why such procedures are not necessary
1. EG _y	Low	These data will be directly used for calculation of emission reductions. Sales record to the grid and other records are used to ensure the consistency. EG _y will be obtained from equation 7 and will be crosschecked with the measurements from M1 and M2, to ensure the conservative criteria. The resulting value of equation 7, supposed to be always less than the measurements of the meters M1 and M2. Please, also refer to procedures explained under D.2.4
1.1 M5	Low	The meter M5 have a 0.2 accuracy class and verification frequency every 2 years.
1.2 M6	Low	The meter M6 have a 0.2 accuracy class and verification frequency every 2 years.
1.3 M3	Low	The meter M3 have a 0.2 accuracy class and verification frequency every 2 years.
1.4 M4	Low	The meter M4 have a 0.2 accuracy class and verification frequency every 2 years.
1.5 M7	Low	The meter M7 have a 0.2 accuracy class and verification frequency every 2 years.
1.6 M8	Low	The meter M8 have a 0.2 accuracy class and verification frequency every 2 years.
1.7 M1	Low	The meter M1 have a 0.2 accuracy class and verification frequency every 2 years.
1.8 M2	Low	The meter M2 have a 0.2 accuracy class and verification frequency every 2 years.
2. EF _y	Low	N/A since this parameter is calculated with data inputs obtained from official and reliable sources.
3. EF _{OM,y}	Low	N/A since this parameter is calculated with data inputs obtained from official and reliable sources.

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4. $EF_{BM,y}$	Low	N/A since this parameter is calculated with data inputs obtained from official and reliable sources.
5. $F_{i,y}$	Low	Data is obtained from official and reliable source reported by CDEC-SIC. Therefore QA/QC procedure is not considered to be necessary.
6. $COEF_i$	Low	N/A since this parameter is calculated with data inputs obtained from official and reliable sources.
7. $GEN_{j/k,y}$	Low	Data is obtained from official CDEC-SIC and CNE databases.
8. Plant name OM	Low	N/A since this parameter is obtained from official and reliable sources.
9. Plant name BM	Low	N/A since this parameter is obtained from official and reliable sources.
10. λ_y	Low	N/A since this parameter is calculated with data inputs obtained from official and reliable sources.
11a. $GEN_{j/k/l,y IMPORTS}$	Low	N/A, in case that imports occurs, data will be obtained directly from reliable and official local source.
11b. $COEF_{i,j,y IMPORTS}$	Low	N/A, in case that imports occurs, data will be obtained directly from reliable and official local source.
12. NCV_i	Low	Data is calculated from the Gross Calorific Value obtained from a local reliable source: The National Energy Balance published by CNE each year. The Gross Calorific Value is then converted into NCV with an approach proposed by the IPCC 2006 guidelines as specified in section D.2.1.1 of the PDD. If such local data (CNE source) are not available or if CNE local data is found to be not reliable (i.e. important variations for a specific year reported NCV value respect the historical average of NCV values published by the same source) for certain monitoring periods during the crediting period, the IPCC default values at the lower limit of the uncertainty at a 95% confidence interval as provided in Table 1.2 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories would be applied instead. It will be considered as a not reliable data when data values obtained from the CNE (local source) represents more than a 20% of variation respect the average values of historical data published by the same source (data values to be used will be compared against data published from previous years by the National Energy Balance).
13. $OXID_i$	Low	Data is obtained from Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories (these are the most conservative values). Any future revision of the IPCC Guidelines should be taken into account.
14. $EF_{CO2,i}$	Low	Data consist in the IPCC default values at the lower limit of the uncertainty at a 95% confidence interval as provided in Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories (these are the most conservative values). Any future revision of the IPCC Guidelines should be taken into account.
15. Hours of operation	Low	This data is obtained directly from the meters at each gen-set and are can be deemed to be very accurate.
16. Fuel Consumption (FCi)	Low	Based on parameters 15 and 19 which can be deemed to be very accurate.

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17. Net calorific value (NCV _{gen-set})	Low	Data is obtained from Chapter1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories (these are the most conservative values). Any future revision of the IPCC Guidelines should be taken into account.
18. Emission factor of fuel (EF _{gen-set})	Low	Data is obtained from Chapter1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories (these are the most conservative values). Any future revision of the IPCC Guidelines should be taken into account.
19. Specific diesel consumption at full load	Low	This data is obtained from manufacturer information and can be deemed to be very accurate.
Others	Low	Default data (for emission factors) and IEA statistics (for energy data) are used to check the local data.

D.4. Please describe the operational and management structure that the project operator will implement in order to monitor emission reduction and any leakage effects, generated by the project activity.

>>

No leakage was identified for this Project activity.

D.5. Name of person/entity determining the monitoring methodology:

>>

The monitoring study was originally concluded in March 2005. The entity determining the baseline as the Carbon Advisor is EcoSecurities Ltd., UK. . The monitoring plan was revised in June 2011 by the project proponent with the help of Perspectives GmbH, Germany¹².

Transmission company.

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**SECTION E. Estimation of GHG emissions by sources****E.1. Estimate of GHG emissions by sources:**

>>

Four diesel gen-sets with a combined installed capacity of 0,801 MW are installed at the power house and the intakes. After commissioning, the gen sets have limited operation time, giving estimated emissions of 460 tCO₂ per year.

Steps to calculate the CO₂ emissions from the operating hours as metered through the hourmeter (steps refer to the table below):

- 1 This value is obtained by multiplying the hours of operation with the Specific fuel consumption
- 2 This value is obtained by multiplying the diesel consumption in liters with the conversion factor from liters to tonnes of diesel, 0.00085 t/Lt, ref. D.S N°66, Chilean Ministry of the Economy)
- 3 This value is obtained by multiplying the Diesel consumption with Diesel NCV (0.0433 TJ/t, ref IPCC, higher value at 95% confidence level).
- 4 This value is obtained by multiplying Net calorific value of the Diesel consumed with the emission factor for diesel (74.8 tCO₂/TJ, ref. IPCC, higher value at 95% confidence level)

			Step No.:					
				1	2	3	4	
			Hour-meter reading	Specific fuel consumption	Diesel consumption	Diesel consumption	Calorific value of diesel consumed	Emissions from fuel
i	Location	MW	Hours	Lt/hr	Lt diesel	t diesel	TJ	tCO ₂
1	La Higuera Power House	0.599	99	134.0	13,266	11.3	0,5	36.52
2	Azufre Intake Center	0.1	6,215	16.5	102,548	87.2	3,7	282.31
3	Tinguiririca Intake Center	0.08	3,576	14.0	50,064	42.6	1,8	137.83
4	Los Helados Intake Center	0.022	74	14.0	1,036	0.9	0,04	2.85
								460

E.2. Estimated leakage:

>>

No leakage was identified

E.3. The sum of E.1 and E.2 representing the project activity emissions:

>>

460 tCO₂ per year

E.4. Estimated anthropogenic emissions by sources of greenhouse gases of the baseline:

>>



Electricity generated emission reduction	Per year	Crediting Period (7 years)
Estimated Simple Adjusted Operating Margin Emissions Factor (EF_{OM_y} in tCO ₂ /MWh)	0.640	-
Build Margin Emissions Factor (EF_{BMy} in tCO ₂ /MWh)	0.361	-
Baseline Emissions Factor (EF_y in tCO ₂ /MWh)	0.501	-
Electricity generated by Project (EG MWh)	811,000	5,677,000
Baseline Emissions (BE tCO ₂)	405,906	2,841,342

E.5. Difference between E.4 and E.3 representing the emission reductions of the project activity:

>>

Given that E.3 is equal to 460 tCO₂, the emission reductions of project activity are equal to 405,446 tCO₂ per year.

E.6. Table providing values obtained when applying formulae above:

Year	Estimation project activity emissions (tonnes of CO ₂ e)	Estimation of baseline emissions (tonnes of CO ₂ e)	Estimation of leakage (tonnes of CO ₂ e)	Estimation of emission reduction (tonnes of CO ₂ e)
2008	460	405,906	0	405,446
2009	460	405,906	0	405,446
2010	460	405,906	0	405,446
2011	460	405,906	0	405,446.
2012	460	405,906	0	405,446
2013	460	405,906	0	405,446
2014	460	405,906	0	405,446.
Total (tonnes of CO ₂ e)	3,220	2,841,342	0	2,838,122

Table 6 - Estimation of emissions during the project activity

**SECTION F. Environmental impacts****F.1. Documentation on the analysis of the environmental impacts, including transboundary impacts:**

>>

The project complies with the specific applicable regulations of the host country in regard to Environment Impact Assessment (EIA). The EIA follows the regulations for EIA System set in Chile by the Supreme Decree N 30/97 of the Ministry General Secretariat of the Presidency, Regulation for the Environmental Impact Assessment System and its modifications set in Supreme Decree N 95/2001, and the Act N 19300 on the Environmental Framework. The EIA has also passed the IFC regulations for EIA, one of the highest standards in the world. La Higuera has received the environmental permit issued by the Chilean government and the approval on environmental matter from the IFC.

The EIA discusses a wide range of environmental impacts related to physical, biotic, human, cultural, patrimonial and landscapes impacts during the plant's construction and operation stages. It identifies the risk or contingency zones and the type of risk associated to them. It also discusses a number of corrective measures and establishes an environmental management plan to deal with the impacts identified. This plan addresses the significant and medium impacts providing measures for their mitigation, restoration or compensation.

There are twelve positive impacts identified with effects on the human environment and the socioeconomic development of the District of San Fernando, where the project is located. The project will allow the generation of clean energy for the region displacing electricity generated from fuel-based power plants. Furthermore, it is going to generate benefits in the form of climate change mitigation.

F.2. If environmental impacts are considered significant by the project participants or the hosts Party, please provide conclusions and all references to support documentation of an environmental impact assessment undertaken in accordance with the procedures as required by the host Party:

>>

During the construction and operation stage, the only significant impact it is related to the loss of habitat for the aquatic fauna and flora. The environmental management plan establishes the measures undertaken to mitigate or compensate the impact. Furthermore, four medium impacts were identified and mitigation measures will be implemented in order to avoid any erosive process and alteration of the landscape after the construction stage.

**SECTION G. Stakeholders' comments****G.1. Brief description how comments by local stakeholders have been invited and compiled:**

>>

According to the Chilean Environmental Law, a project of the size and characteristics of La Higuera is required to develop an Environmental Impact Study (EIS). La Higuera carried out the EIS between December 2001 and December 2003, with the obligation to carry public community consultations through publications in local newspapers and public hearings at a community level.

The public hearings were done with the local authorities and the local community. More than 100 people attended the presentation. Many of them were representatives of local interested groups. Both events took place in San Fernando, VI Region, in March 16, 2004.

1. Local authorities: More than 15 different authorities were present, including all relevant services dealing with the environmental and sectoral authorizations of the project. The presentation was done in 2 steps. One related to the technical and environmental aspect of the project and the second one related to the CDM characteristics of the project.

2. Local community: in accordance with the EIS procedure, the community was invited by the environmental authorities at a gathering, in which the project developer would present the major characteristics of the project.

G.2. Summary of the comments received:

>>

Local authorities: Questions were raised related to the road conditions and how the project would benefit the community. With respect to more work places and environmental conditions, all comments were positive, recognizing the benefits for the community of the project and the low impact on the environment.

With respect to the CDM aspect of the presentation, the authorities present showed a lot of interest, asking for more details related to this topic, especially with respect to the procedures and applicability conditions.

Local community: With respect to the technical and environmental aspects of the presentation, similar questions and comments were raised, as in the previous presentation. In relation to the CDM aspects, not big interested was shown by the audience, except for one question that questioned the emission reduction of the project by emphasizing that regardless of this project, no future thermal plants are going to be displaced due to the constant increase in energy needs.

G.3. Report on how due account was taken of any comments received:

>>

Although no major issues were raised that could be related to the environmental or CDM aspect of the project, all comments and questions were duly taken into account by the project developer. The main concern of the community was related to the improvements of roads, concern that the project developer resolved by a private agreement with the local authorities. All other technical and environmental aspects were resolved at the EIS and approved by the environmental authorities.



Annex 1

CONTACT INFORMATION ON PARTICIPANTS IN THE PROJECT ACTIVITY

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Annex 2
INFORMATION REGARDING PUBLIC FUNDING

There is no public funding used in this project.



Annex 3
BASELINE INFORMATION

EX-ANTE SIMPLE ADJUSTED OM EMISSION FACTOR

The following tables summarize the input data values used for the Simple adjusted OM Calculation for 2010:

Table 1. Factors of Oxidation, CO₂ emission and Net Calorific Value used for COEF_i estimation:

Fuel type i	OXID _i	EF _{CO₂,i} tCO ₂ /GJ	NCVi (GJ/kg; GJ/m ³ for Natural Gas)	COEF _i (tonnesCO ₂ /kg or tonnesCO ₂ /m ³ for gas fuels)
Coal	1	0.08950	0.02784	0.002491879
Diesel	1	0.07260	0.04335	0.003147523
Natural Gas	1	0.05430	0.03520	0.001911252
Liquified Natural Gas	1	0.05430	0.03520	0.001911252
IFO 180	1	0.07550	0.04176	0.003153131

Table 2. Fossil fuel consumption

Fuel Type i	Unit	Fuel consumption
Coal	Kg/year	3,250,840,000
Diesel	Kg/year	845,090,000
IFO 180	Kg/year	620,000
Liquified Natural Gas	m ³ /year	1,144,250,000
Natural Gas	m ³ /year	266,330,000

Table 3. Electricity generation per source

Source	Generation (MWh/year)
Coal	8,768,895
Biomass	765,421
Biogas	7,649
Diesel	4,624,949
Hydro (reservoir)	13,207,626
Hydro (run off river)	7,982,898
IFO 180	1,069
Liquified Natural Gas	5,867,171
Natural Gas	495,378
Wind	318,519
no data about the source*	1,153,500
Total (including CDM power units)	43,193,074
Electricity Generated by CDM power units	1,919,407
TOTAL electricity considered in OM calculation (excluding CDM power units)	41,273,667

***If for a power plant there is no data about the energy source then as a conservative approach it is considered as a “j” (fossil fuel) source with an emission factor equal to 0. Electricity produced by power units from registered CDM project activities were excluded from the OM calculation.**

Table 4. $EF_{OM, simple-adjusted, y}$ calculation

Parameter	Unit	Value
Low-cost/must-run sources electricity generation	MWh/year	20,369,505
Lamda (λ_{2010})	%	0.005594
Hours of the year when low cost mus trun are on the margin	Hours	49
$\sum F_{i,j,y} * COEF_{i,j} / \sum GEN_{i,y}$	tCO ₂ /MWh	0.643821
$\sum F_{i,k,y} * COEF_{i,k} / \sum GEN_{k,y}$	tCO ₂ /MWh	0.000000
$EF_{OM, simple-adjusted, y}$	tCO ₂ /MWh	0.640220

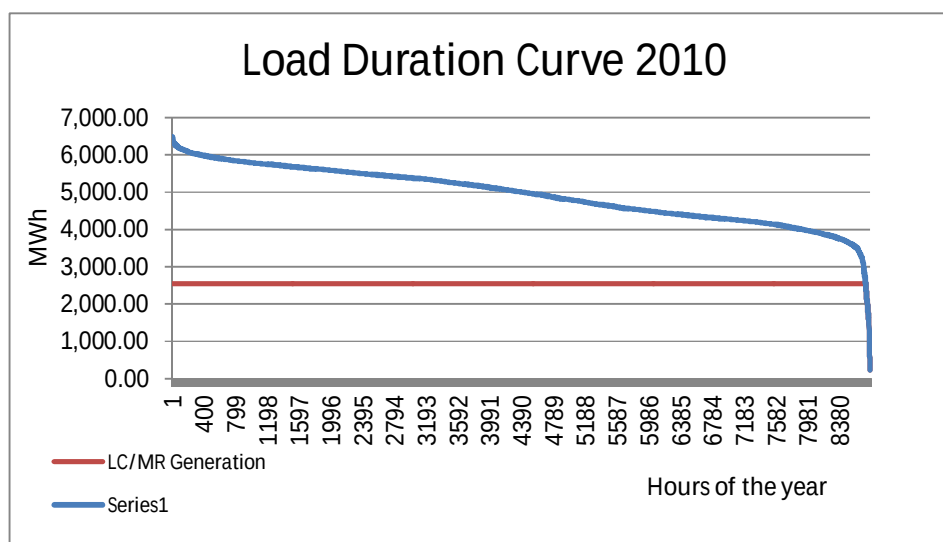


Table 1 Lamda curve

BUILD MARGIN CALCULATIONS FOR LA HIGUERA HYDRO

From the ACM0002 the build margin calculation is defined as:

Option 1. Calculate the Build Margin emission factor $EF_{BM,y}$ ex ante based on the most recent information available on plants already built for sample group m at the time of PDD submission. The sample group m consists of either:

- *the five power plants that have been built most recently, or*
- *the power plants capacity additions in the electricity system that comprise 20% of the system generation (in MWh) and that have been built most recently.*

Project participants should use from these two options that sample group that comprises the larger annual generation.



In the SIC network, 20% of annual generation is approximately 7,000,000MWh. Using the above methodology the build margin will be calculated using the most recent available data for the past 12 months (Dec 2003 to November 2004) and estimation of annual energy generation where complete data is not yet available. The results for this calculation are shown in the table below

Plant	Year	Fuel Type	2004 Energy Generated (MWh)	Annual CO ₂ Emissions (tonnes)
Licanten	2004	Biomass	21,391	0
Valdivia	2004	Biomass	155,707	0
Anihue TG	2004	Diesel	870	3180.104
Horcones TG	2004	Diesel	12,078	45475.49
TG Coronel	2004	Diesel	0	0
Itata	2004	Biomass	0	0
Ralco	2004	Hydro	1,332,259	0
Nehuenco II	2003	CCGT	1,996,332	736178.5
Cholguan	2003	Biomass	93,234	0
Nehuenco 9B Diesel	2002	Diesel	39,223	31801.04
Nehuenco 9B CCGT	2002	CCGT	68,377	43170.96
SF de Mostazal	2002	Diesel	9,380	9219.121
Chacabuguito	2002	Hydro	152,945	0
Taital 2 Diesel	2000	Diesel	59	50.88166
Taital 2 OCGT	2000	OCGT	364,308	244256.8
Taital 1	2000	OCGT	624,493	418758.3
Peuchen	2000	Hydro	262,257	0
Mampil	2000	Hydro	174,152	0
Petropower	1998	Coal	526,232	474286.1
Nehuenco Diesel	1998	Diesel	47,800	24486.8
Nehuenco	1998	CCGT	1,800,051	740268.4
Totals			7,681,148	2,771,133
Build Margin			0.361	

Table 5 - Data used to calculate the build margin



Annex 4 MONITORING PLAN

Backup gen-sets

There are four diesel gen-sets as a backup system which will provide electricity to the essential consumption and auxiliary services. The largest gen-set is located in the Power House fuelled by an external tank of 3000 liter diesel. The other gen-set are located in the intakes centre and they are mainly used for the management of the gates of the intakes.

The gen-sets will operate in case of emergencies, when power system maintenance (lines and substations) is on-going and in case energy from the SIC (grid) is not available.

The gen-set in Power House can provide enough energy for a black start of one unit allowing the start-up of the other one and then recover the sub-system. This capability is called autonomous start-up.

It is expected that the operational cases are going to be very specific and sporadic, since, maintenance are scheduled carefully in advance and the external grid is very stable; hence this black start (after a black out) are uncommon.

It is a common practice of the O&M team to have records of maintenance and operation of all plant equipment including the gen-sets.

The gen-sets will serve as back up. These gen-sets will run in case of emergencies and if there is a total power outage in the plant.

The features of each gen-sets are:

Mark	Model	Serial number	Place	Power Stand-by	i	Voltage	Frequency	Speed
Battistella	MAQ51P 27A	1464001	Power House	599kVA	1	380/220V	50 Hz	1500 rpm
Lureye	J110K	J110K09013646	Azufre Intake Center	100kVA	2	400V	50 Hz	1500 rpm
Lureye	J88K	J88K09000807	Tinguiririca Intake Center	80kVA	3	380/220V	50 Hz	1500 rpm
Olympian	GEP22-4	OLY00000KD4J 04648	Los Helados Intake Center	22kVA	4	380/220V	50 Hz	1500 rpm

The gen-sets with 0.599 MW, 0.080 MW, 0.100 MW and 0.022 MW of installed capacity, will only be used for that specific purpose and will not supply power to the grid. The gen-sets have been used in the past for the construction of the plant and it is expect that during plant operation, these will be used only in emergencies. It is estimated that their emissions in normal operation of the power plant will be well below 1% of the average emission reduction of the project activity Nevertheless, La Higuera will use the hourmeters of the gen-sets to monitor the use of the engines . Related emissions will be considered in the monitoring reports.

Emissions of each gen-set is obtained through the calculation of excise duties and the calorific value of the diesel that deals with each one. The default values of NCV and Emission factor of the diesel will be derived from the IPCC, higher value at 95% confidence level.



In case of a failure of the hour meters and/or in the absence of sufficient data recording it is assumed for such periods that the gen-sets have been operated continuously at full load as a conservative approach but capped to the maximum possible operating hours according to available and proper meter readings.